

DAX Exercise 1

Introduction to DAX in Power BI

Data Analytics | Power BI Desktop | Beginner Level

Duration	Level	Tool	Dataset
45-60 min	Beginner	Power BI Desktop	Sales + Products

1. Overview

DAX (Data Analysis Expressions) is the formula language built into Power BI. You use DAX to create calculations that go beyond what you loaded in Power Query — things like running totals, percentage contributions, conditional logic, and time comparisons. In this exercise you will write your first DAX formulas, understand the difference between measures and calculated columns, and build a set of business metrics from a sales dataset.

What You Will Learn

- Understand the difference between a Measure and a Calculated Column
- Write basic aggregation measures: SUM, COUNT, AVERAGE
- Use DIVIDE for safe division (avoiding divide-by-zero errors)
- Use CALCULATE to apply filters to a measure
- Write an IF calculated column for conditional logic
- Use simple date functions: YEAR, MONTH
- View your measures in a simple card and table visual

Prerequisites

Before starting this exercise you should be comfortable with:

- Opening Power BI Desktop and loading CSV files (see Power Query Exercise 1)
- Navigating the Power BI interface: Report view, Data view, Model view
- Basic concepts of tables, columns, and rows

The Datasets

This exercise uses two CSV files that have a relationship — just like two tables in a database.

File	Column	Description
Sales_DAX.csv	SaleID	Unique ID for each transaction
	Date	Date of sale (MM/DD/YYYY)
	CustomerName	Purchasing company
	SalesRep	Name of the sales representative
	ProductID	Links to the Products table
	Quantity	Units sold
	UnitPrice	Price per unit at time of sale
	Discount	Discount rate (0.00 – 0.20)
Products_DAX.csv	ProductID	Unique product identifier (links to Sales)
	ProductName	Product display name
	Category	Electronics / Office Supply / Furniture
	CostPrice	Wholesale cost per unit
	UnitPrice	Standard retail price

2. Setup — Load the Data

Step 1 — Load Both CSV Files

1. Open Power BI Desktop.
2. Click Get data > Text/CSV. Navigate to Sales_DAX.csv and click Transform Data.
3. In Power Query Editor, rename the query from Sales_DAX to Sales (double-click the name in the Queries pane).
4. Set the data types: Date → Date, Quantity → Whole Number, UnitPrice → Decimal Number, Discount → Decimal Number.
5. Click Home > Close & Apply.
6. Repeat steps 2–5 for Products_DAX.csv, renaming the query to Products and setting CostPrice and UnitPrice to Decimal Number.

Step 2 — Create the Relationship

Power BI needs to know how the two tables are connected so DAX formulas can cross-reference them.

1. Click the Model view icon on the left navigation bar (looks like three connected boxes).

2. You should see both the Sales and Products tables. If Power BI has already drawn a line between them on the ProductID column, the relationship was auto-detected — you can skip to step 4.
3. If no relationship exists, drag the ProductID column from the Products table and drop it onto the ProductID column in the Sales table. A relationship line appears.
4. Double-click the relationship line to verify: Many-side = Sales (ProductID), One-side = Products (ProductID). Click OK.

One-to-Many Relationships

Products table: one row per product (the 'one' side, also called the lookup/dimension table).

Sales table: many rows can share the same ProductID (the 'many' side, also called the fact table).

This relationship lets DAX look up product details (like Category or CostPrice) from any row in the Sales table.

Step 3 — Add a Revenue Column in Power Query (Preparation)

Before writing DAX, go back to Power Query to add a Revenue calculated column to the Sales table. This is the base figure your DAX measures will aggregate.

1. In the Home ribbon click Transform data to reopen Power Query Editor.
2. Select the Sales query. Click Add Column > Custom Column.
3. Name the column Revenue and enter the formula:

Power Query — Revenue Column

```
= [Quantity] * [UnitPrice] * (1 - [Discount])
```

1. Set the Revenue column type to Decimal Number.
2. Click Close & Apply.

Expected Result

Model view shows a relationship line between Sales[ProductID] and Products[ProductID].

Sales table has 100 rows with a Revenue column.

Products table has 15 rows.

3. DAX Concepts — Measures vs Calculated Columns

Before writing any DAX, it is essential to understand the two places you can use it.

	Calculated Column	Measure
Where stored	Inside a table — adds a new column	Displayed in the Fields pane — no new column
When calculated	Once, row by row, at data load time	On demand, when used in a visual
Evaluated at	Row context (knows the current row)	Filter context (knows the current visual filters)
Good for	Categorising rows, flags, label combinations	Totals, averages, %, KPIs that respond to slicers
Example	= IF([Revenue] > 500, 'High', 'Low')	= SUM(Sales[Revenue])

Golden Rule

If the result belongs to a ROW (one value per transaction) → use a Calculated Column.

If the result is a SUMMARY or KPI (one value for the whole report or a filter selection) → use a Measure.

Most DAX you write in practice will be Measures.

How to Create a Measure

1. In the Data view or Report view, select the table the measure belongs to (e.g. Sales) in the Fields pane.
2. In the Home ribbon click New Measure (or right-click the table name > New measure).
3. A formula bar opens. Type your DAX formula and press Enter or click the tick mark.
4. The measure appears in the Fields pane under the chosen table with a calculator icon.

How to Create a Calculated Column

1. In the Data view, select the table you want to add a column to.
2. In the Home ribbon click New Column (or right-click the table header > New column).
3. Type your DAX formula in the formula bar and press Enter.
4. A new column appears in the table with the values calculated for every row.

4. Task 1 — Total Revenue (SUM)

SUM is the most common DAX function. It adds up all values in a column, respecting whatever filters are active in the current visual.

Function	Syntax
SUM	SUM(<column>) — adds all values in the column

Instructions

1. Select the Sales table in the Fields pane.
2. Click New Measure in the Home ribbon.
3. Type the formula below and press Enter.
4. Rename the measure Total Revenue if Power BI did not pick up the name automatically.

DAX Measure — Total Revenue

```
Total Revenue = SUM(Sales[Revenue])
```

Test Your Measure

Go to Report view. From the Visualisations pane choose the Card visual.

Drag Total Revenue from the Fields pane onto the card.

The card should display the sum of all revenue across all 100 transactions.

Expected Result

Total Revenue card displays approximately \$142,000 – \$160,000 (exact value depends on data).

The measure appears in the Fields pane under Sales with a calculator icon (Σ).

5. Task 2 — Transaction Count & Customer Count

Counting rows and counting distinct values are two different operations — and both are essential for business reporting.

Function	What It Does
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COUNTROWS(<table>)	Counts the number of rows in a table
DISTINCTCOUNT(<column>)	Counts unique values in a column — ignores duplicates

Task 2A — Total Transactions

1. Create a new measure in the Sales table.
2. Enter the formula below and press Enter.

DAX Measure — Total Transactions

Total Transactions = COUNTROWS(Sales)

Task 2B — Total Customers

1. Create a new measure in the Sales table.
2. Enter the formula below and press Enter.

DAX Measure — Total Customers

Total Customers = DISTINCTCOUNT(Sales[CustomerName])

Test Both Measures

Add two more Card visuals to the report canvas and drag each measure onto one card. Try adding a table visual with Category (from Products) and both measures — notice how the counts update per category.

Expected Result

Total Transactions = 100

Total Customers = 10 (there are 10 unique customers in the dataset)

When filtered by category in a table visual the counts reduce to show only rows matching that category.

6. Task 3 — Average Sale & Revenue per Transaction

Function	What It Does
AVERAGE(<column>)	Returns the arithmetic mean of a numeric column
DIVIDE(<numerator>, <denominator>, [alternateResult])	Divides two values. Returns 0 (or your chosen value) instead of an error when the denominator is 0.

Task 3A — Average Revenue per Sale

1. Create a new measure in the Sales table.
2. Enter the formula below and press Enter.

DAX Measure — Average Revenue per Sale

```
Avg Revenue per Sale = AVERAGE(Sales[Revenue])
```

Task 3B — Revenue per Transaction (using DIVIDE)

This measure calculates the same average a different way — by dividing Total Revenue by Total Transactions. Notice how it references two other measures. This is a key pattern in DAX: building measures from measures.

1. Create a new measure in the Sales table.
2. Enter the formula below and press Enter.

DAX Measure — Revenue per Transaction

```
Revenue per Transaction = DIVIDE([Total Revenue], [Total Transactions], 0)
```

Why DIVIDE Instead of / ?

Using / in DAX will throw an error if the denominator is zero.

DIVIDE(a, b, 0) safely returns 0 when b = 0 — preventing report errors.

The third argument is the alternate result — you can use BLANK() to return nothing instead of 0.

Expected Result

Avg Revenue per Sale ≈ \$1,400 – \$1,600 (average over 100 rows)

Revenue per Transaction should be very close to Avg Revenue per Sale.

If you filter the visuals by a single sales rep the values will update automatically.

7. Task 4 — CALCULATE (Filtered Measures)

CALCULATE is the most important function in DAX. It evaluates a measure or expression in a modified filter context — letting you ask questions like 'what is the revenue for Electronics only?' even when the visual shows all categories.

CALCULATE Syntax

```
CALCULATE( <expression>, [filter1], [filter2], ... )
```

expression — the measure or formula to evaluate

filter1... — one or more filter conditions that override the current context

Task 4A — Electronics Revenue

1. Create a new measure in the Sales table.
2. Enter the formula below and press Enter.

DAX Measure — Electronics Revenue

```
Electronics Revenue =  
    CALCULATE(  
        [Total Revenue],  
        Products[Category] = 'Electronics'  
    )
```

Task 4B — Furniture Revenue

1. Create a new measure in the Sales table.
2. Enter the formula below.

DAX Measure — Furniture Revenue

```
Furniture Revenue =  
    CALCULATE(  
        [Total Revenue],  
        Products[Category] = 'Furniture'  
    )
```


Task 4C — Electronics % of Total

Now combine CALCULATE and DIVIDE to express Electronics as a percentage of all revenue.
Note: CALCULATE with ALL removes existing filters — explained in the box below.

DAX Measure — Electronics % of Total

```
Electronics % of Total =  
    DIVIDE(  
        [Electronics Revenue],  
        CALCULATE([Total Revenue], ALL(Products)),  
        0  
    ) * 100
```

What Does ALL() Do?

ALL(<table or column>) removes all filters from the specified table or column.

Without ALL, the denominator would also be filtered by Electronics, making the result always 100%.

With ALL(Products), the denominator is always the grand total regardless of slicers.

This pattern — DIVIDE(filtered, CALCULATE(total, ALL(...))) — is the standard way to calculate % of total in DAX.

Expected Result

Electronics Revenue: approximately \$40,000 – \$55,000

Furniture Revenue: approximately \$55,000 – \$75,000

Electronics % of Total: approximately 25% – 35%

Test: add a slicer on Category. Electronics Revenue and Electronics % of Total should remain unchanged — they ignore the slicer because CALCULATE overrides the filter context.

8. Task 5 — IF Calculated Column

Use IF in a calculated column to assign a label to every row based on a condition. Unlike a measure, this runs row by row and stores the result permanently in the table.

IF Syntax

```
IF( <logical_test>, <value_if_true>, [value_if_false] )
```

Task 5A — Sale Size Flag

Flag each transaction as Large, Medium, or Small based on the Revenue value.

1. In Data view, select the Sales table.
2. Click New Column in the Home ribbon.
3. Enter the formula below and press Enter.

DAX Calculated Column — Sale Size

```
Sale Size =  
    IF(  
        Sales[Revenue] >= 500,  
        'Large',  
        IF(  
            Sales[Revenue] >= 100,  
            'Medium',  
            'Small'  
        )  
    )
```

Task 5B — Discount Flag

Add a second calculated column that flags whether a discount was applied.

1. Create a new column in the Sales table.
2. Enter the formula below and press Enter.

DAX Calculated Column — Discount Applied

```
Discount Applied =  
    IF(  
        Sales[Discount] > 0,  
        'Yes',  
        'No'  
    )
```

Test Your Columns

Go to Data view and click the Sales table. You should see the two new columns at the right. In Report view, add a table visual with Sale Size and Discount Applied as rows. Add Total Revenue as a value.

You can now see revenue broken down by discount status and sale size.

Expected Result

Sale Size column: every row has 'Large', 'Medium', or 'Small'.

Discount Applied column: every row has 'Yes' or 'No'.

Approximately 40–50% of rows should show Discount Applied = Yes.

9. Task 6 — Date Columns (YEAR, MONTH)

DAX has many date functions. In this task you will add two simple calculated columns that extract the year and month from the Date column — useful for grouping and time-based visuals.

Function	What It Returns
YEAR(<date>)	Integer year (e.g. 2024)
MONTH(<date>)	Integer month number 1–12
FORMAT(<value>, <format_string>)	Converts a value to formatted text (e.g. month name)

Task 6A — Sale Year

1. Create a new column in the Sales table.
2. Enter the formula below.

DAX Calculated Column — Sale Year

```
Sale Year = YEAR(Sales[Date])
```

Task 6B — Sale Month Number

1. Create a new column in the Sales table.
2. Enter the formula below.

DAX Calculated Column — Sale Month

```
Sale Month = MONTH(Sales[Date])
```

Task 6C — Month Name

1. Create a new column in the Sales table.
2. Enter the formula below.

DAX Calculated Column — Month Name

```
Month Name =  
    FORMAT(Sales[Date], 'MMMM')
```

Using Date Columns in Visuals

Add a clustered bar chart. Place Month Name on the X axis and Total Revenue on the Y axis. Notice the bars may not sort chronologically — month names sort alphabetically by default. To fix: select Month Name column in Data view > Column Tools ribbon > Sort by Column > Sale Month.

Expected Result

Sale Year column: all values should be 2024.
Sale Month column: integer values 1 through 6 (January to June).
Month Name column: values like 'January', 'February', ... 'June'.
Bar chart shows 6 bars sorted January → June.

10. Task 7 — Profit and Profit Margin

Now that the relationship between Sales and Products is set up, you can reference columns from the Products table inside a measure on the Sales table. This is where relationships pay off.

Task 7A — Total Cost

Total Cost is the sum of (Quantity × CostPrice) for every transaction. Because CostPrice lives in the Products table and Quantity lives in Sales, you use SUMX — a row-by-row iterator — to multiply them together before summing.

SUMX Syntax

```
SUMX( <table>, <expression> )
```

SUMX iterates through every row of a table, evaluates the expression for each row, then sums all results.

1. Create a new measure in the Sales table.
2. Enter the formula below.

DAX Measure — Total Cost

```
Total Cost =  
    SUMX(  
        Sales,  
        Sales[Quantity] * RELATED(Products[CostPrice])  
    )
```

What is RELATED?

RELATED(<column>) fetches a value from the related (Products) table for the current row in the Sales table.

It only works inside row-context functions like SUMX, or inside Calculated Columns.

It travels from the many-side (Sales) to the one-side (Products) along the relationship you created earlier.

Task 7B — Gross Profit

1. Create a new measure in the Sales table.
2. Enter the formula below.

DAX Measure — Gross Profit

```
Gross Profit = [Total Revenue] - [Total Cost]
```

Task 7C — Profit Margin %

1. Create a new measure in the Sales table.
2. Enter the formula below.

DAX Measure — Profit Margin %

```
Profit Margin % =  
    DIVIDE([Gross Profit], [Total Revenue], 0) * 100
```

Test Profit Measures

Add a table visual with Category, Total Revenue, Total Cost, Gross Profit, and Profit Margin %.

Add a row for each category — you should see Electronics, Furniture, and Office Supply. Office Supply products have the highest margins (low cost, moderate price).

Expected Result

Total Cost: approximately \$60,000 – \$80,000

Gross Profit: approximately \$60,000 – \$90,000

Profit Margin %: approximately 40% – 55% overall

Office Supply Profit Margin % is the highest (around 60%+), Furniture is lowest.

11. Solutions — All DAX Formulas

Below is the complete set of measures and calculated columns for this exercise. In Power BI you can verify your own formulas by clicking a measure or column in the Fields pane and checking the formula bar.

All Measures — Sales Table

Complete Measures

```
// Task 1 – Aggregation
Total Revenue = SUM(Sales[Revenue])

// Task 2 – Counts
Total Transactions = COUNTROWS(Sales)
Total Customers = DISTINCTCOUNT(Sales[CustomerName])

// Task 3 – Averages
Avg Revenue per Sale = AVERAGE(Sales[Revenue])
Revenue per Transaction = DIVIDE([Total Revenue], [Total Transactions], 0)

// Task 4 – CALCULATE with filters
Electronics Revenue =
    CALCULATE([Total Revenue], Products[Category] = 'Electronics')

Furniture Revenue =
    CALCULATE([Total Revenue], Products[Category] = 'Furniture')

Electronics % of Total =
    DIVIDE(
        [Electronics Revenue],
        CALCULATE([Total Revenue], ALL(Products)),
        0
    ) * 100

// Task 7 – Profit
Total Cost =
    SUMX(Sales, Sales[Quantity] * RELATED(Products[CostPrice]))

Gross Profit = [Total Revenue] - [Total Cost]

Profit Margin % =
    DIVIDE([Gross Profit], [Total Revenue], 0) * 100
```

All Calculated Columns — Sales Table

Complete Calculated Columns

```
// Task 5A — Sale Size flag
Sale Size =
    IF(Sales[Revenue] >= 500, 'Large',
        IF(Sales[Revenue] >= 100, 'Medium', 'Small'))

// Task 5B — Discount flag
Discount Applied =
    IF(Sales[Discount] > 0, 'Yes', 'No')

// Task 6A — Year
Sale Year = YEAR(Sales[Date])

// Task 6B — Month number
Sale Month = MONTH(Sales[Date])

// Task 6C — Month name (set Sort by Column = Sale Month)
Month Name = FORMAT(Sales[Date], 'MMMM')
```

Key Function Summary

Function	Type	What It Does
SUM(column)	Aggregation	Adds all values in a column
COUNTROWS(table)	Aggregation	Counts total rows in a table
DISTINCTCOUNT(column)	Aggregation	Counts unique values
AVERAGE(column)	Aggregation	Arithmetic mean of a column
DIVIDE(a, b, alt)	Math	Safe division — returns alt if b = 0
CALCULATE(expr, filter)	Filter	Evaluates expr with modified filter context
ALL(table/column)	Filter modifier	Removes all filters from a table or column
IF(test, true, false)	Logical	Returns one of two values based on a condition
SUMX(table, expr)	Iterator	Row-by-row multiply then sum
RELATED(column)	Relationship	Fetches a value from the related table
YEAR / MONTH(date)	Date	Extracts year or month number
FORMAT(value, fmt)	Text	Converts a value to formatted text

12. Reflection Questions

Answer the following questions based on your work in this exercise.

1. In your own words, explain the difference between a Measure and a Calculated Column. Give one example of when you would choose each.

2. In Task 4, why did you need to use ALL(Products) inside the denominator of Electronics % of Total? What would happen to the result if you removed ALL(Products)?

3. Why is DIVIDE(a, b, 0) safer than writing a / b directly in DAX? In what business scenario would a zero denominator realistically occur?

4. SUMX iterates row by row before summing. Why couldn't you just write SUM(Sales[Quantity] * Products[CostPrice]) instead of using SUMX and RELATED?

5. You created Month Name and Sale Month columns. Why is it necessary to have the numeric Sale Month column even though you only display Month Name on visuals?

6. A colleague adds a Category slicer to the report page. How will each of your measures behave when a single category is selected? Which measures will change, and which will stay the same? Why?

Suggested Answers

1. Measure vs Calculated Column

A Calculated Column runs once per row at data load and stores the result in the table — use it when each row needs its own label or derived value (e.g. Sale Size per transaction). A Measure is computed on demand, responding to filters and slicers — use it for KPIs and totals shown in visuals (e.g. Total Revenue).

2. Why ALL(Products) in the denominator

Without ALL, CALCULATE inherits the current filter context from the visual. If the visual is already filtered to Electronics, both the numerator and denominator would only count Electronics rows, always returning 100%. ALL(Products) forces the denominator to always sum across all categories, giving a true percentage of the total.

3. DIVIDE vs /

If the denominator is zero, the / operator raises an error that can break visuals. DIVIDE returns 0 (or BLANK) instead of an error. A realistic zero-denominator scenario: a new sales rep with no transactions yet — Total Transactions = 0, so Revenue per Transaction would divide by zero.

4. Why SUMX and RELATED instead of SUM

SUM works on a single column. You cannot multiply two columns from different tables inside SUM. SUMX iterates through every Sales row and, for each row, uses RELATED to look up the CostPrice from the related Products row — then sums all the per-row products. Without RELATED there is no row context to cross the relationship.

5. Why keep Sale Month as a number

Month names are text and sort alphabetically (April, August, December ...) rather than chronologically. Storing Sale Month as an integer (1–12) and using Sort by Column tells Power BI the correct sort order for Month Name without changing what users see in the visual.

6. Slicer behaviour

Most measures — Total Revenue, Total Transactions, Total Cost, Gross Profit, Avg Revenue per Sale — will change because they respect the filter context imposed by the slicer. Electronics Revenue, Furniture Revenue, and Electronics % of Total will remain fixed because their CALCULATE filters override the slicer context (or use ALL to ignore it).

13. Challenge Extension

Completed all tasks? Push your DAX knowledge further with these challenges.

A	Top Sales Rep Revenue Create a measure that returns the revenue of the top-performing sales rep using: <code>Top Rep Revenue = MAXX(VALUES(Sales[SalesRep]), [Total Revenue])</code> Add a Card visual to display it.
	Office Supply % of Total Using the pattern from Task 4C, write a measure that shows Office Supply revenue as a % of total. Then write a third measure: Furniture % of Total. Verify that Electronics % + Office Supply % + Furniture % $\approx$ 100%.
C	Large Sale Count Write a measure that counts only the transactions flagged as Sale Size = 'Large': <code>Large Sale Count = CALCULATE([Total Transactions], Sales[Sale Size] = 'Large')</code> Combine with a measure for Large Sale Revenue.
	Discounted Revenue Write a measure that sums revenue only for rows where a discount was applied: <code>Discounted Revenue = CALCULATE([Total Revenue], Sales[Discount] > 0)</code> Then calculate what % of total revenue came from discounted sales.
E	Monthly Revenue Trend Create a line chart with Month Name on the X axis (sorted by Sale Month) and Total Revenue on the Y axis. Add a constant reference line at the average monthly revenue. Which months are above average and which are below?

Challenge Solutions

Challenge A–D DAX Solutions

```
// Challenge A – Top Sales Rep Revenue
Top Rep Revenue = MAXX(VALUES(Sales[SalesRep]), [Total Revenue])

// Challenge B – Office Supply % of Total
Office Supply % of Total =
    DIVIDE(
        CALCULATE([Total Revenue], Products[Category] = 'Office Supply'),
```

```

        CALCULATE([Total Revenue], ALL(Products)),
        0
    ) * 100

Furniture % of Total =
    DIVIDE(
        CALCULATE([Total Revenue], Products[Category] = 'Furniture'),
        CALCULATE([Total Revenue], ALL(Products)),
        0
    ) * 100

// Challenge C – Large Sale Count & Revenue
Large Sale Count = CALCULATE([Total Transactions], Sales[Sale Size] = 'Large')
Large Sale Revenue = CALCULATE([Total Revenue], Sales[Sale Size] = 'Large')

// Challenge D – Discounted Revenue & %
Discounted Revenue = CALCULATE([Total Revenue], Sales[Discount] > 0)
Discounted % of Total =
    DIVIDE([Discounted Revenue], [Total Revenue], 0) * 100

```